

LONDON BICYCLE ANALYSIS PROJECT

TASK :1 At what station did the bike trip with `rental_id` 57635395 end?

 **LONDON BYCYCLE ANALYSIS** Run Download

```
1 SELECT end_station_name
2 FROM `bigquery-public-data.london_bicycles.cycle_hire`
3 WHERE rental_id = 57635395
```

 This query will process 2.89 GB when run.

Using on-demand processing quota

Query results

Job information		Results	Visualization	JSON	Execution details
Row	end_station_name				
1	East Village, Queen Elizabeth Ol...				

Key insight : The analysis reveals bike trip with `rental_id` 57635395 end at east village ,queen elizabeth olympic park.

Task 2 : how many bike trips lasted for 20 minutes or longer?

 **Untitled query** Run Download Share Schedule Open in More Save

```
1 SELECT duration,start_station_name
2 FROM `bigquery-public-data.london_bicycles.cycle_hire`
3 WHERE duration >= 1200;
```

 Query completed

Using on-demand processing quota

Query results

[Chat](#) [Save results](#)

Job information		Results	Visualization	JSON	Execution details	Execution graph
Row	duration	start_station_name				
1	14220	Mallory Street, Marylebone				

Results per page: 50 1 – 50 of 26441016

Key insight : total trip count : a total of 26441016 trips were identified that lasted for more than 20 or more minutes

Data observation : the result shows various starting points across london where users rented bikes and it also shows their duration in seconds.

Task 3: What is the name of the station whose `start_station_id` is 111?

Untitled query [Run](#) [Download](#) [Share](#) [Schedule](#) [Open in](#) [More](#) [Save](#)

```
1 SELECT start_station_name
2 FROM `bigquery-public-data.london_bicycles.cycle_hire`
3 WHERE start_station_id = 111 ;
```

✓ Query completed
Using on-demand processing quota

Query results [Chat](#) [Save results](#)

Job information **Results** Visualization JSON Execution details Execution graph

Row	start_station_name
1	Park Lane , Hyde Park

Results per page: 50 1 – 50 of 276216

Key insight :

trip volume : a total of 276216 bicycle trips started from this location

Observation : the high number of trip suggest the “ hyde park “ is one of the most popular starting points for bicycle rentals in london

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```
1 SELECT start_station_id, rental_id, start_station_name
2 FROM `bigquery-public-data.london_bicycles.cycle_hire`
3 WHERE bike_id = 1710
```

✓ Query completed
Using on-demand processing quota

Query results [Chat](#) [Save results](#)

Job information **Results** Visualization JSON Execution details Execution graph

Row	start_station_id	rental_id	start_station_name
9	568	75808940	Bishop's Bridge Road West, Bay...

Results per page: 50 1 – 50 of 4543

Task 4: Return all the `rental_ids`, station IDs, and station names that `bike_id` 1710 started from.

Key insight: the specific bike has completed a total of 4543 trips in recorded data set

Usage pattern : the high number of trips indicates that the individual bike has served thousands of commuters across various station, it shows bike is in very good condition and used by many people everyday. And its very popular across different stations.

Task 5 : What is the `bike_model` of `bike_id` 58782?

The screenshot shows the Google BigQuery interface. At the top, there's a toolbar with buttons for 'Run', 'Download', 'Share', 'Schedule', 'Open in', 'More', and 'Save'. Below the toolbar, the SQL query is entered in a text area:

```
1 SELECT bike_model
2 FROM `bigquery-public-data.london_bicycles.cycle_hire`
3 WHERE bike_id = 58782
```

Below the query, a status bar indicates 'Query completed' and 'Using on-demand processing quota'. The 'Query results' section is visible, showing a table with one row and one column:

Row	bike_model
1	CLASSIC

At the bottom right, it says 'Results per page: 50' and '1 - 50 of 429'.

Key insight : this query identifies exactly the kind of bike model is which is “classic” being used for specific bike id

Observation : the query returned 429 results which represent the total number of trips made by this single bike across different dates and stations

Task 6 : Top five baby names for boys in the United States in 2014.

The screenshot shows the Google BigQuery interface. At the top, there's a toolbar with buttons for 'Run', 'Save', 'Download', 'Share', 'Schedule', and 'C'. Below the toolbar, the SQL query is entered in a text area:

```
1 SELECT name, count
2 FROM `my-project-11021-495506.babynames.names_2014`
3 WHERE gender = 'M'
4 ORDER BY count DESC LIMIT 5
```

Below the query, a status bar indicates 'This query will process 622.98 KB when run.' and 'Using on-demand processing quota'. The 'Query results' section is visible, showing a table with two columns:

Row	name	count
1	Noah	19316

At the bottom right, it says 'Results per page: 50' and '1 - 5 of 5'.

KEY INSIGHT : Used sql to query and identify the top 5 male baby names of 2014 demonstrating proficiency in data filtering , sorting, and ranking techniques

BABY NAMES PROJECT

TASK 1 : FIND TOP 5 BABY BOY NAMES FROM YEAR 2015-2019

Untitled query

Run

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Schedule

Open in

```
1 SELECT NAME,
2 SUM(COUNT) AS TOTAL_COUNT
3 FROM `my-project-11021-495506.BABY_NAMES_PROJECT_2015_2019.names_2015_2019`
4 WHERE
5     GENDER = 'M'
6 GROUP BY NAME
7 ORDER BY TOTAL_COUNT DESC
8 LIMIT 5
```

Only saved queries can be shared

Query results

Chat Save results Open in

Job information

Results

Visualization

JSON

Execution details

Execution graph

row	NAME	TOTAL_COUNT
1	Liam	95854
2	Noah	94634
3	William	74830
4	James	70715
5	Mason	69340

TASK 2: FINDING AVERAGE DIAMETER OF TREES IN 2015

tree_ce... 995

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Untitled query

Run

Save

Download

Share

Schedule

```
1 SELECT AVG(tree_dbh) AS tree_diameter
2 FROM `bigquery-public-data.new_york_trees.tree_census_2015`
3 LIMIT 1000
```

Query completed

Using on-demand processing quota

Query results

Chat Save results Open in

Job information

Results

Visualization

JSON

Execution details

Execution graph

row	tree_diameter
1	11.27978701000...